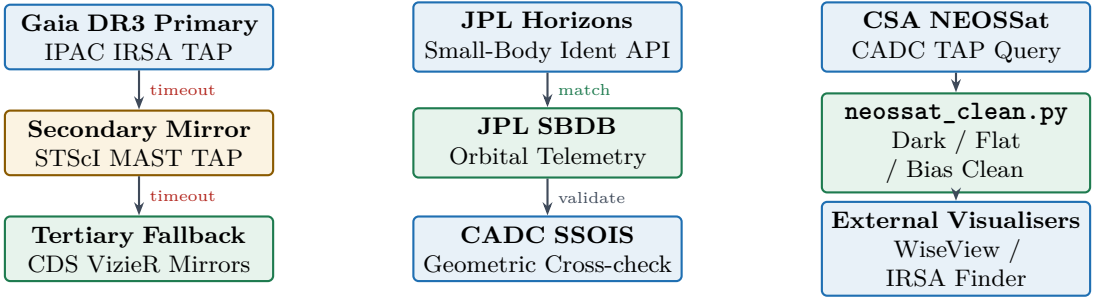


Dual Ingestion Modes Compared:

Feature	Cached Mode (`data_cache/`)	Live Streaming Mode (`live_ingest.py`)
Throughput	Instant replay from SSD / GCS snapshots	4 concurrent worker threads parsing HTTP stream in RAM
Disk Storage	Few hundred MBs of FITS files on disk	**Zero bytes written to disk** (streamed into `BytesIO`)
External Dependency	Offline-ready; independent of archive status	Live dependence on IRSA SIA2 latency and availability
Use Case	Scheduled batch jobs, evaluation, ML training	On-demand ad-hoc exploration of any sky coordinate